

ECOMIX LLP · EASYMIX · SELECTIVE ZONAL ISOLATION

NANOCEM UT-9

An ultra-fine injection grout for selective zonal isolation. A D95 of 9 µm or finer penetrates the microfractures and pores that ordinary oil-well cement cannot reach.

PARTICLE FINENESS

D95 ≤ 9 µm

D50 ~3.5 µm

28-DAY STRENGTH

65–75 MPa

grade M500

TREATMENTS COMPLETED

7 wells

January — September 2026

REFERENCE RESULT

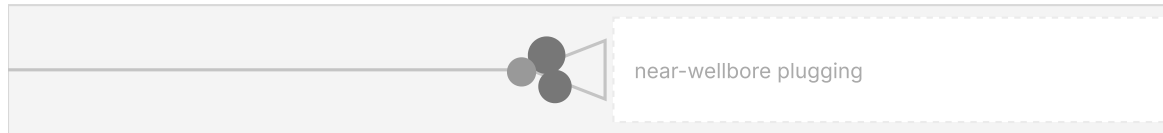
1.1 → 16.4

t/day on well 343

FINENESS DECIDES IT, NOT PUMP PRESSURE

A squeeze job is decided by the penetrating ability of the slurry.

ORDINARY CEMENT · 40–80 μm



NANOCEM UT-9 · $D_{95} \leq 9 \mu\text{m}$



As it is pumped, the slurry gives up its water to the formation and the solid phase travels towards the filtration channels. If the particle size is comparable to the channel aperture, the near-wellbore zone is plugged instead: a filter cake builds on the wall, nothing but filtrate goes further, and the specific injectivity of the interval falls without the channel itself being filled. Ordinary oil-well cement at 40–80 μm runs into that limit at any pump pressure.

A fineness of $D_{50} \sim 3.5 \mu\text{m}$ and $D_{95} \leq 9 \mu\text{m}$ gives the slurry a penetrating ability approaching that of a solids-free system: part of the volume enters the microfractures and the matrix of the near-wellbore reservoir rock, the rest builds set cement in the perforation channels. After WOC the barrier sits within the interval, not on the casing wall.

PRODUCT DATA SHEET

Dry blend, grey powder · supplied in 25 kg and 1000 kg packs

Fineness D50 / D95	~3.5 µm / ≤ 9 µm
Dry product density	2.95 g/cm³
Slurry density	1.85–1.95 g/cm³
Working life of the slurry	4–5 hours
Initial set	~60 min
Full set	24 hours at +25 °C
Strength after 24 hours	≥ 12 MPa

Strength after 28 days	65–75 MPa · grade M500
Bond to concrete	≥ 2.5 MPa
Water resistance	W14–W16
Layer thickness	5–100 mm
Chemical resistance	oil, brines, sulphates
Shelf life	12 months

WHAT NANOCEM UT-9 ISOLATES

Three uses proven on a producing well stock.

USE 01

Remedial squeeze work

Selectively shutting off spent perforation intervals. The oil-bearing part of the reservoir stays on production — the well returns at a higher rate.

USE 02

Eliminating crossflow behind casing

Cutting off the channels behind the production casing that connect the horizons above and below. The fineness fills narrow gaps that ordinary oil-well cement cannot reach.

USE 03

Restoring casing integrity

Filling cracks and micro-gaps in the cement sheath and restoring zonal isolation. A bond of 2.5 MPa or better grips the previously set cement.

HOW IT IS DONE AT THE WELL

The established squeeze sequence — as run on well 357, September 2026

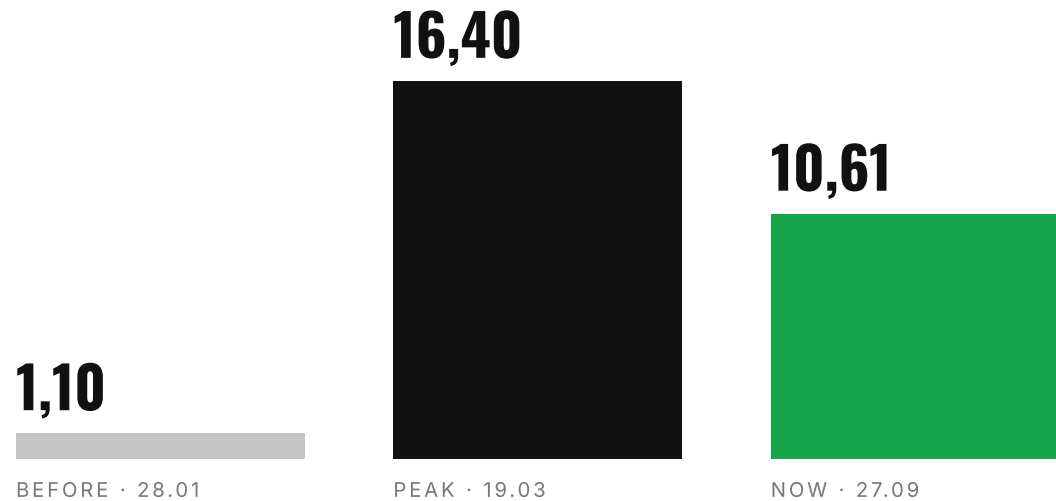
01 Preparation Shut in, log, identify the inflow intervals and the specific injectivity. The slurry is mixed at a W/C ratio of about 0.87 — deliberately mobile, or the solid phase will not enter the channels it is chosen for.	02 A three-stage squeeze First a portion with the annulus open — the grout is distributed across all the intervals. Then the tubing is pulled and a second portion goes in. Only then is the packer set and the rest squeezed in closed.	03 Injectivity as the control On the closed-in stage the pressure builds and then holds: the specific injectivity of the interval has fallen to zero and the formation has stopped taking slurry. While the channels are open, pressure does not build at a constant pump rate.
04 Waiting on cement under pressure The well stays under pressure throughout — the cement gains strength under compression rather than simply standing in the hole.	05 Drill-out and pressure test The bridge is drilled out and each interval is tested separately as it is opened, so a leaking interval is located immediately.	06 Perforation and start-up Logging, perforation, ramp-up. The effect is judged from the oil-rate trend over two to three weeks.

SEVEN TREATMENTS ON A PRODUCING WELL STOCK

Clastic reservoir, 620–760 m · daily operating logs as of 27.09.2026

WELL	ENTRIES	ISOLATION OUTCOME	DATE	RESULT
343	1	● Tight on the 1st	Jan 2026	1.10 → 16.40 t/day at the peak · ×9.6 over baseline, eighth month
301	1	● Tight on the 1st	06.06.2026	10.28 t/day — a steady regime, no stoppages
303	1	● Tight on the 1st	13.07.2026	2.86 t/day — the well's record over the whole record
342	2	● Tight on the 2nd	10.05 → 31.05	7.86 t/day · ×13.3 over a baseline of 0.59 t/day
311	1	● Confirmed after ramp-up	09.07.2026	6.31 t/day — the rate tripled after ramp-up
357	1	● Tight on pressure test	24.09.2026	direct test: 3 of 3 intervals hold
305	1	● Cannot be assessed — target changed	18.07.2026	switched to a shallower horizon in September

WELL 343



Treated in January 2026. The oil rate rose almost fifteenfold — from 1.10 to 16.40 t/day at the peak. The effect is holding into its eighth month.

Oil rate, t/day. Eighth month after the treatment — the effect is holding.

PRODUCED SINCE THE JOB

≈3,033 t
 over 235 days on stream

GAIN OVER BASELINE

×9.6
 1.10 → 10.61 t/day

CURRENT RATE

10.61 t/day
 as of 27.09.2026

EFFECT DURATION

8 months
 and continuing

AN HONEST LIMIT OF APPLICABILITY

Isolating the perforations delivers its full effect only where the cement sheath behind the casing is intact.

Logging reports on two edge wells show poor cement behind casing: continuous contact over only 6.25% and 30.4% of the hole. With contact like that, inflow bypasses the isolated perforations through channels behind the casing, so closing the holes does not produce a full seal — and the gain in rate is only partial.

INDICATOR

Poor cement behind casing

Continuous contact over less than a third of the hole on the bond log. The sheath must be assessed before the job, or the result will be partial.

INDICATOR

An edge position in the pool

The central-eastern core seals on the first entry. At the edges a second entry or a different scheme is required.

WHAT FOLLOWS

Candidate selection

Candidates should be picked by position and casing condition. Operating figures on their own do not predict the outcome — the sheath must be assessed by bond log before the job.

TIGHTNESS CAN BE PROVEN STRAIGHT AWAY

On well 357 the isolation was confirmed by direct test — before start-up, without waiting for production statistics.

Cement bridge	40 atm · 15 min	tight
Intervals 657–663 and 669.5–672 m	30 atm	tight
Interval 679–684 m	40 atm · 15 min	tight

The testing was done in stages as the bridge was drilled out: each interval was checked as soon as it was opened. Testing the whole hole at once would not locate a leaking interval. A caveat: the test was at 30–40 atm against a squeeze pressure of 80 atm, and it says nothing about crossflow behind the casing.

WHAT WE OFFER

The cement, a proven squeeze procedure and transparent analytics on every treatment.

01

The cement

An ultra-fine blend of our own manufacture. Supplied in 25 kg and 1000 kg packs, 12-month shelf life.

02

The procedure

A three-stage squeeze with pressure control, a separate pressure test on each interval and subsequent re-perforation.

03

The analytics

Daily monitoring of the well stock, effect measured from the oil-rate trend, failures analysed. All of it open at nanocem.app.

NANOCEM.APP — THE WELL STOCK, THE TIGHTNESS REVIEW AND THE WORK PROGRAMME